$$\text{This implies, } 4x^3 + x^2\left(3y^2\frac{dy}{dx}\right) + (2x)y^3 - 5y^4\frac{dy}{dx} = 2 + 0$$

$$\text{This implies, } 4x^3 + 3x^2y^2\frac{dy}{dx} + 2xy^3 - 5y^4\frac{dy}{dx} = 2$$

$$\text{This implies, } (3x^2y^2 - 5y^4)\frac{dy}{dx} = 2 - 4x^3 - 2xy^3$$

$$\text{or } \frac{dy}{dx} = \frac{2 - 4x^3 - 2xy^3}{3x^2y^2 - 5y^4}.$$

Example 10.20

Find $\frac{dy}{dx}$ if $\sin y = y\cos 2x$.

Solution

We have $\sin y = y\cos 2x$.

$$\text{Differentiating, } \frac{d}{dx}\sin y = \frac{d}{dx}(y\cos 2x)$$

$$\text{That is, } \cos y\frac{dy}{dx} = y(-2\sin 2x) + \cos 2x\frac{dy}{dx}$$

$$\text{This implies, } (\cos y - \cos 2x)\frac{dy}{dx} = -2y\sin 2x$$

$$\text{or } \frac{dy}{dx} = \frac{-2y\sin 2x}{\cos y - \cos 2x}.$$

10.4.4 Logarithmic Differentiation

By using the rules for differentiation and the table of derivatives of the basic elementary functions, we can now find automatically the derivatives of any elementary function, except for one type, the simplest representative of which is the function $y = x^x$. Such functions are described as power-exponential and include, in general, any function written as a power whose base and index both depend on the independent variable.

In order to find by the general rules the derivative of the power-exponential function $y = x^x$, we take logarithms on both sides to get

$\log y = x \log x, x > 0$

Since this is an identity, the derivative of the left-hand side must be equal to the derivative of the right, we obtain by differentiating with respect to x (keeping in mind the fact that the left hand side is a function of function) :

$$\frac{1}{y}\frac{dy}{dx} = \log x + 1$$

$$\text{Hence } \frac{dy}{dx} = y(\log x + 1) = x^x(\log x + 1)$$

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